Does finer Δt stabilise the integration?

exp061 · 11 July 2026 · Draft

Abstract

exp060 showed the free signed-recurrent ceiling trains on the Spiking Heidelberg Digits (SHD) spoken-digit classification benchmark [1] but diverges intermittently: scattered epochs return NaN (not-a-number), the pre-clip gradient norm spikes into the millions, and the excitatory-to-excitatory (E→E) recurrent weight grows without bound. Here E and I denote the excitatory and inhibitory populations, and "signed" means the recurrent weights are free to take either sign (no Dale's law). A NaN appeared at **epoch 2, when the weights were still small**, so the divergence is not weight-growth runaway. The program plan parks that observation as its first hypothesis: the NaN is exponential-Euler **integration stiffness** at the coarse Δt = 1.0 ms (the numerical error a large integration timestep introduces into a stiff differential equation), so halving then quartering Δt should drop the NaN-epoch rate toward zero.

This experiment sweeps Δt ∈ {1.0, 0.5, 0.25} ms on a single seed (42), holding exp060′s Rung A recipe otherwise fixed, and measures the plan's three registered stability metrics against Δt. The hypothesis is refuted: not one of the three improves as Δt shrinks. The NaN-epoch rate does not fall, the peak pre-clip gradient norm climbs orders of magnitude in the wrong direction, and the kill criterion fires, so coarse-Δt stiffness is not the cause and Δt leaves the recipe.

Methods

This holds exp060′s Rung A recipe (its free signed-recurrent configuration) fixed and varies only the integration timestep, Δt ∈ {1.0, 0.5, 0.25} ms, on a single seed (42): a fast first read rather than an error-barred result. It measures the plan's three registered stability metrics against Δt:

- the **NaN-epoch rate**, the fraction of epochs with a non-finite loss or a NaN forward pass;
- the **max pre-clip gradient norm**, the peak global gradient magnitude before clipping, the quantity that spiked to millions in exp060;
- the **max recurrent-weight norm**, the peak $\|W_{\mathrm{ee}}\|$, the E→E block that grew unbounded.

Parameter	Value
Dataset	SHD, 1000-sample subset
Model	COBANet (conductance-based spiking network), 256 hidden units, all four recurrent blocks trainable, signed (no Dale's law)
Swept variable	$\Delta t \in \{1.0, 0.5, 0.25\}$ ms at trial duration $T = 1000$ ms, so the number of integration steps $T_{\mathrm{steps}} \in \{1000, 2000, 4000\}$
Seeds	1 (seed 42), a quick first read
Training	30 epochs, learning rate 0.001, weight decay 0.001, batch size 32, surrogate-gradient backpropagation through time (BPTT)
Regulariser	upper firing-rate bound (threshold $\theta_u = 100$, strength $s_u = 0.06$), load-bearing, from exp060
Compute	runpod

Δt is the **only** thing that varies across the three runs; everything else is exp060′s recipe verbatim, so any change in the stability metrics is attributable to Δt alone. Δt = 1.0 reproduces exp060′s exact failure setting, so the finer-Δt arms are read against a grounded baseline rather than an abstract one.

Kill criterion. If NaN persists at $\Delta t = 0.25$ (the finest step), coarse integration is not the cause, so drop Δt from the recipe and look upstream (a forward-pass state clamp).

Results

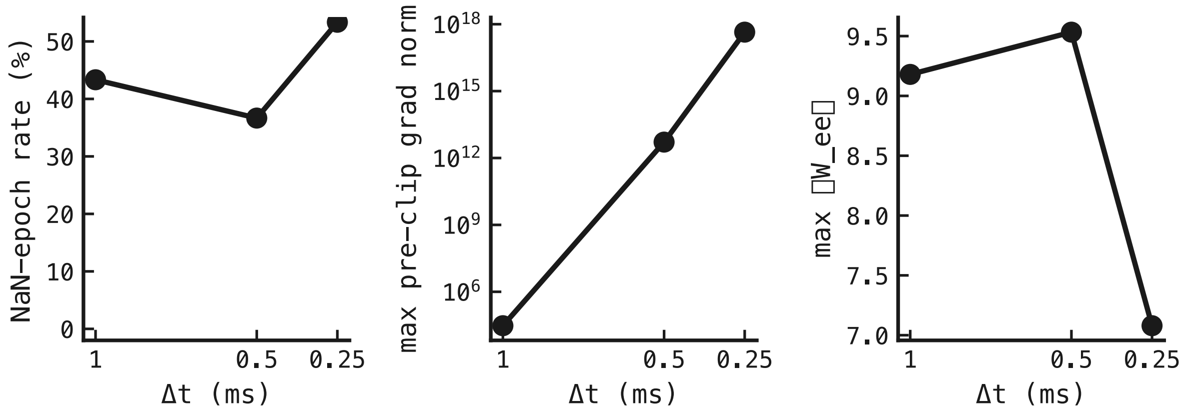

Figure 1: The three stability metrics against Δt in ms (coarse $\rightarrow$ fine, left $\rightarrow$ right on each panel): the NaN-epoch rate (%, left), the peak pre-clip gradient norm (log scale, centre), and the peak E$\rightarrow$E recurrent-weight norm $\|W_{ee}\|$ (right). **What we expect if the plan's hypothesis holds:** all three fall as Δt shrinks, the NaN-epoch rate toward zero and the gradient and weight norms toward bounded values. **What we see refutes it.** Not one of the three metrics improves as Δt shrinks. The NaN-epoch rate does not fall toward zero: 13 of 30 epochs diverge at $\Delta t =$ 1.0, 11 at 0.5, and 16 at 0.25 (flat-to-worse, not a descent). The peak pre-clip gradient norm moves the wrong way and violently, climbing from $\approx 10^4$ at $\Delta t = 1.0$ to $\approx 10^{18}$ at $\Delta t = 0.25$, roughly 14 orders of magnitude the wrong way. Only the excitatory-to-excitatory recurrent-weight norm $\|W_{ee}\|$ stays in one range (7.1–9.5), so this is not a runaway-weight story.

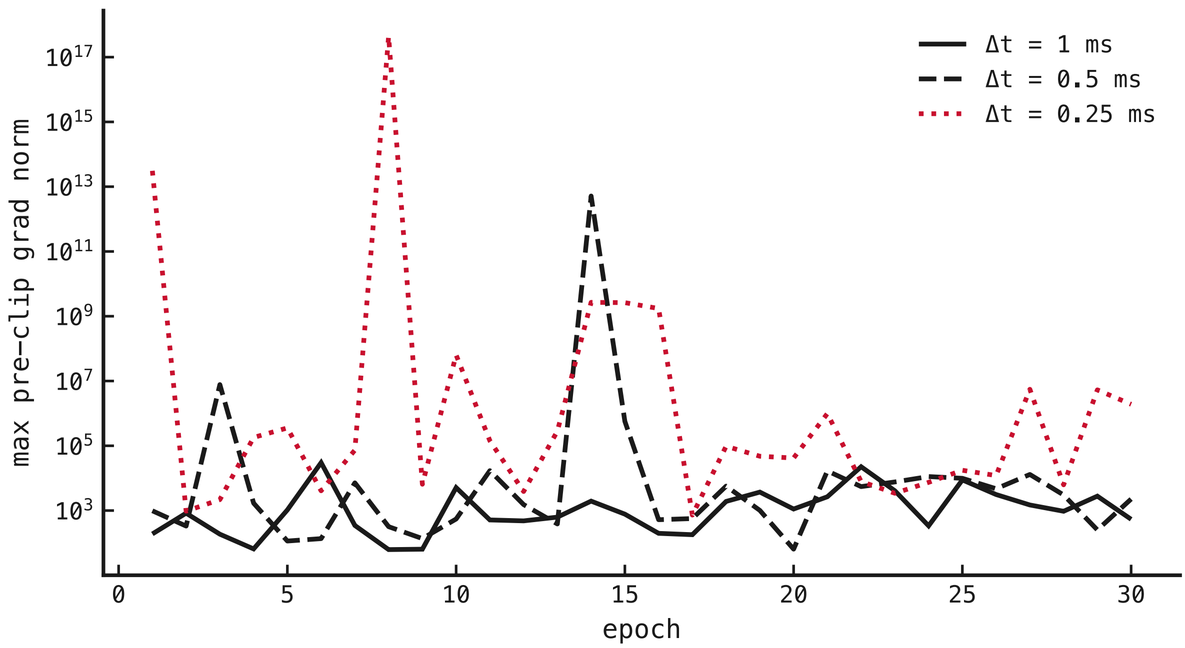

Figure 2: The max pre-clip gradient norm per epoch (x-axis epoch, y-axis log scale), one line per Δt. This shows **when** the gradient spikes fire, not just their peak: a stiff integration diverges on individual forward passes rather than drifting up smoothly.

The coarse-Δt run reproduces exp060′s divergence (13 of 30 epochs return NaN, 12 NaN forward passes), so the mechanism is genuinely exercised; this is not a null run. But finer Δt does not remove it: at $\Delta t = 0.25$ the NaN-epoch rate is 16 of 30, no lower than at $\Delta t = 1.0$, and the peak pre-clip gradient norm is **larger** at every halving, reaching $\approx 10^{18}$ at the finest step.

The kill criterion fires. The plan pre-registered: *if NaN persists at $\Delta t = 0.25$, coarse integration is not the cause, so drop Δt from the recipe and look upstream.* It persists. So the divergence is not exponential-Euler stiffness at a coarse timestep. The mechanism is visible in the sweep itself: quartering Δt quarters the step but **quadruples** the BPTT unroll (1000 $\rightarrow$ 4000 steps), and the longer gradient chain through the signed E$\rightarrow$I$\rightarrow$E recurrence explodes far worse than any stiffness a finer step relieves. Δt is not the stabiliser; it leaves the recipe.

Where this points. With neither Δt (here) nor weight size (exp060: a NaN at epoch 2 with tiny weights) explaining the divergence, the plan's reserved fallback, a forward-pass state clamp bounding voltage and conductance, becomes the live lead. The reduced-scale CPU cross-check agreed directionally: no NaN at 128 samples, but the same non-monotonic gradient blow-up peaking at $\Delta t = 0.25$.

Caveat. Single seed, so these are point estimates, not error bars. The effect is large (orders of magnitude, not margins), so a seed sweep is unlikely to reverse the direction; the plan's three-seed default is the confirming run.

References

1. B. Cramer, Y. Stradmann, J. Schemmel & F. Zenke (2020). The Heidelberg spiking data sets for the systematic evaluation of spiking neural networks. *IEEE Transactions on Neural Networks and Learning Systems* 33(7):2744–2757. doi:10.1109/TNNLS.2020.3044364